SIMATS ENGINEERING

ASSESSMENT TOOL 2-Short Answer Test

COURSE NAME: Data Structures **COURSE CODE: CSA0310**

PRESENTED BY:

192521004 -BHUVANESH. S

Graphs and Graph Algorithms:

1. What is a graph without cycles called?

A graph without any cycles is called an acyclic graph.

If it is undirected, it is commonly called a forest.

2. Which graph representation uses a two-dimensional array?

An adjacency matrix uses a two-dimensional array.

It stores the connection between every pair of vertices.

3. What type of graph models one-way road systems?

A directed graph is used to represent one-way roads.

The direction of an edge represents the allowed direction of travel.

4. Which graph model is used to represent dependencies?

A directed graph is commonly used to represent dependencies.
An edge from A to B can mean that A must be completed before B.

5. Which graph type is commonly used in job assignment problems?

A bipartite graph is commonly used for job assignment problems.
One set represents workers and the other set represents jobs.

6. For which type of graph does topological ordering exist?

Topological ordering exists for a Directed Acyclic Graph (DAG).
It arranges vertices so that every prerequisite comes before its dependent vertex.

7. Which data structure is used to store results in DFS-based topological sorting?

A stack is used to store vertices in DFS-based topological sorting.
Vertices are pushed after all their adjacent vertices are processed.

8. Which graph model is suitable for course prerequisite systems?

A Directed Acyclic Graph (DAG) is suitable for course prerequisites.
Vertices represent courses and directed edges represent prerequisite relationships.

9. What type of graph is used in compiler build systems?

A Directed Acyclic Graph (DAG) can represent dependencies in compiler build systems.

It ensures that required components are built before dependent components.

10. Which shortest path algorithm can detect negative cycles?

The Bellman-Ford algorithm can detect negative-weight cycles.

It is useful when a graph contains negative edge weights.

Answer Key:

Q. No.	Answer
1	Acyclic graph
2	Adjacency matrix
3	Directed graph
4	Directed graph
5	Bipartite graph
6	Directed Acyclic Graph (DAG)
7	Stack
8	Directed Acyclic Graph (DAG)
9	Directed Acyclic Graph (DAG)
10	Bellman-Ford algorithm